CodeCraft Code Cheat Sheet

For students using Code mode in CodeCraft.

1. Basic Pattern

```
var button1 = new Button('page1_button1');
page.add(button1);

button1.onClick(() => {
  console.log('Button clicked');
});
```

2. Rules To Remember

- Every element starts with new ClassName('element_id')
- Show an element on screen with page.add(element)
- If an element is not added with page.add(...), it will not show in Run mode
- Parent Container elements are auto-added behind the scenes when needed
- Add click logic with .onClick(() => { ... })
- Use page.goTo('page2') to move to another page

3. Page Object

page.add(element)

Adds an element to the current page.

```
var heading1 = new Heading('page1_heading1');
page.add(heading1);
```

page.goTo(pageIdOrPageName)

Moves to another page by page id or page name.

```
button1.onClick(() => {
  page.goTo('page2');
});
```

You can also use a page name:

```
page.goTo('Page 2');
```

4. Console

Use the debug console while learning:

```
console.log('hello');
console.warn('check this');
console.error('something went wrong');
```

5. Common Element Properties

Many elements support these easy properties:

- .text
- .value
- .label
- .content
- .src

-   .title
-   .placeholder

Example:

```
    var input1 = new Input('page1_input1');
    page.add(input1);

    console.log(input1.value);
```

6. Input And Output Example

```
    var number1 = new Input('page1_input1');
    var number2 = new Input('page1_input2');
    var result = new Label('page1_label1');
    var addButton = new Button('page1_button1');

    page.add(number1);
    page.add(number2);
    page.add(result);
    page.add(addButton);

    addButton.onClick(() => {
      result.content = String(Number(number1.value) + Number(number2.value));
    });
```

7. Element Classes

Layout

Toolbar

```
    var toolbar1 = new Toolbar('page1_toolbar1');
    page.add(toolbar1);
```

Used for app title bars.

Card

```
    var card1 = new Card('page1_card1');
    page.add(card1);
```

Used for grouped content.

List

```
    var list1 = new List('page1_list1');
    page.add(list1);
```

Used for item lists.

Segment

```
    var segment1 = new Segment('page1_segment1');
    page.add(segment1);
```

Used for switching between options.

Separator

```
    var separator1 = new Separator('page1_separator1');
    page.add(separator1);
```

Used for divider lines.

Spacer

```
    var spacer1 = new Spacer('page1_spacer1');
    page.add(spacer1);
```

Used to add empty vertical space.

Container

```
    var container1 = new Container('page1_container1');
```

Usually you do not need to add Container in student code because
CodeCraft auto-adds parent containers for layout.

Input

Button

```
    var button1 = new Button('page1_button1');
    page.add(button1);
```

Use .onClick(...) for actions.

FAB

```
    var fab1 = new FAB('page1_floatingactionbutton1');
    page.add(fab1);
```

Floating Action Button.

Searchbar

```
    var searchbar1 = new Searchbar('page1_searchbar1');
    page.add(searchbar1);
```

LabelledInput

```
    var labelledInput1 = new LabelledInput('page1_labelledinput1');
    page.add(labelledInput1);
```

Input

```
    var input1 = new Input('page1_input1');
    page.add(input1);
```

Select

```
    var select1 = new Select('page1_select1');
    page.add(select1);
```

Range

```
    var range1 = new Range('page1_rangeinput1');
    page.add(range1);
```

DatePicker

```
    var date1 = new DatePicker('page1_datetimepicker1');
    page.add(date1);
```

Checkbox

```
    var checkbox1 = new Checkbox('page1_checkbox1');
    page.add(checkbox1);
```

Radio

```
    var radio1 = new Radio('page1_radiobutton1');
    page.add(radio1);
```

Toggle

```
    var toggle1 = new Toggle('page1_toggle1');
    page.add(toggle1);
```

Display

Heading

```
    var heading1 = new Heading('page1_heading1');
    page.add(heading1);
```

Label

```
    var label1 = new Label('page1_label1');
    page.add(label1);
```

Good for short text output.

Textarea

```
    var textarea1 = new Textarea('page1_textbox1');
    page.add(textarea1);
```

Design name is Textarea, internal element type is based on the textbox component.

Badge

```
    var badge1 = new Badge('page1_badge1');
    page.add(badge1);
```

Chip

```
    var chip1 = new Chip('page1_chip1');
    page.add(chip1);
```

Spinner

```
    var spinner1 = new Spinner('page1_spinner1');
    page.add(spinner1);
```

Image

```
    var image1 = new Image('page1_image1');
    page.add(image1);
```

Link

```
    var link1 = new Link('page1_link1');
    page.add(link1);
```

Video

```
var video1 = new Video('page1_video1');
page.add(video1);
```

Icon

This element exists in Design mode, but is not available as a runtime class in Code mode right now.

8. Simple Navigation Example

```
var nextButton = new Button('page1_button1');
page.add(nextButton);

nextButton.onClick(() => {
  page.goTo('page2');
});
```

9. Simple Form Example

```
var nameInput = new Input('page1_input1');
var submitButton = new Button('page1_button1');
var resultLabel = new Label('page1_label1');

page.add(nameInput);
page.add(submitButton);
page.add(resultLabel);

submitButton.onClick(() => {
  resultLabel.content = 'Hello ' + nameInput.value;
});
```

10. Quick Memory Tips

- new ClassName('id') creates the element
- page.add(...) shows it
- .onClick(...) gives it behavior
- .value reads user input
- page.goTo(...) changes page
- console.log(...) helps debug